BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Timed Paper 2 Practice

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. In a file-size calculation, should the final answer include units?

2. Q = (A AND B) OR C. A=1, B=1, C=0. What is Q?
   A) 1
   B) 0
   C) 2
   D) 10

3. "A translator is needed only to change the program's comments into another human language." Correct this. Explain why.

4. Merge the sorted lists [2, 7] and [4, 6] into ascending order. Give the values in order. Why not just join the lists?

Today's objectives

Explain the core ideas of timed paper 2 practice using accurate OCR terminology. Apply a clear method to the worked and guided questions.

Key vocabulary - Read, cover, recall

Paper 2 timing
The OCR J277/02 paper has 80 marks in 90 minutes.

Code tracing
Write a reliable value table before selecting the output.

Program design
Plan input, processing, selection, iteration and output.

Error spotting
Check indexes, loop bounds, equality tests and branching.

BLACK PEN: Write one definition in your own words.

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Learn and practise

New Learning - Key ideas

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Plan input, processing, selection, iteration and output.

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I Do - Worked model

Trace count=0; for i=1 to 5; if i MOD 2=0 then count=count+1. Output count.

- Loop through i=1,2,3,4,5.
- Only i=2 and 4 satisfy even test.
- Count progresses 0,1,1,2,2.
- Final output is 2.

We Do - Guided practice

1. Design a three-line correction to a failed validation check using OR.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Count the 1 bits in a five-bit binary value using iteration and selection.

```
array bits = [1,0,1,1,0]
ones = 0
for i = 0 to 4
if bits[i] == 1 then
ones = ones + 1
endif
next i
print(ones)
```

You Do - Independent exam questions

Q2. State what an accumulator stores. [2 marks]

Q3. Write a valid condition for 0<=score<=100. [2 marks]

Q4. Explain one common index error in zero-based arrays. [3 marks]

Exit Ticket - True or False?

A trace table helps follow loop values and decisions.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. Yes; show a calculation and label the final unit, such as bits or bytes.

2. A - 1

3. A CPU runs machine instructions. High-level source code must be translated before or as it runs.

4. (a) [2, 4, 6, 7]. Keep taking the smaller next item. Just joining the lists would put 7 before 4.

I Do - Explain the model

Loop through i=1,2,3,4,5.

Only i=2 and 4 satisfy even test.

Count progresses 0,1,1,2,2.

Final output is 2.

We Do - Model answers

1. Check endpoints.

Code First - Answer

The original five-bit number 10110 has three 1 bits, so output is 3. Changing index 1 to 1 gives four 1 bits (11110). The original binary value is 16+4+2 = 22 denary; counting set bits and converting place values are different operations.

You Do - Self-mark

Q2: A running total updated as a program executes.

Q3: score >= 0 AND score <= 100.

Q4: Using length instead of length·1 as final index can access outside array bounds.

GREEN PEN: Choose the hardest answer. Correct it on lined paper.